EasyVisa Project

Business Presentation

Contents

Executive Summary

The profile of the applicants for whom the visa status can be approved:

Primary information to look at:

- Education level - At least has a Bachelor's degree - Master's and doctorate are preferred.
- Job Experience - Should have some job experience.
- Prevailing wage - The median prevailing wage of the employees for whom the visa got certified is around 72k.

Secondary information to look at:

- Unit of Wage - Applicants having a yearly unit of wage.
- Continent - Ideally the nationality and ethnicity of an applicant shouldn't matter to work in a country but previously it has been observed that applicants from Europe, Africa, and Asia have higher chances of visa certification.
- Region of employment - Analysis shows that Midwest region have more chances of visa approval.

Executive Summary

The profile of the applicants for whom the visa status can be denied:

Primary information to look at:

- Education level - Doesn't have any degree and has completed high school.
- Job Experience - Doesn't have any job experience.
- Prevailing wage - The median prevailing wage of the employees for whom the visa got certified is around 65k.

Secondary information to look at:

- Unit of Wage - Applicants having an hourly unit of wage.
- Continent - Ideally the nationality and ethnicity of an applicant shouldn't matter to work in a country but previously it has been observed that applicants from South America, North America, and Oceania have higher chances of visa applications getting denied.

Executive Summary

- Additional information of employers and employees can be collected to gain better insights. Information such as:
    - Employers: Information about the wage they are offering to the applicant, Sector in which company operates in, etc
    - Employee's: Specialization in their educational degree, Number of years of experience, etc

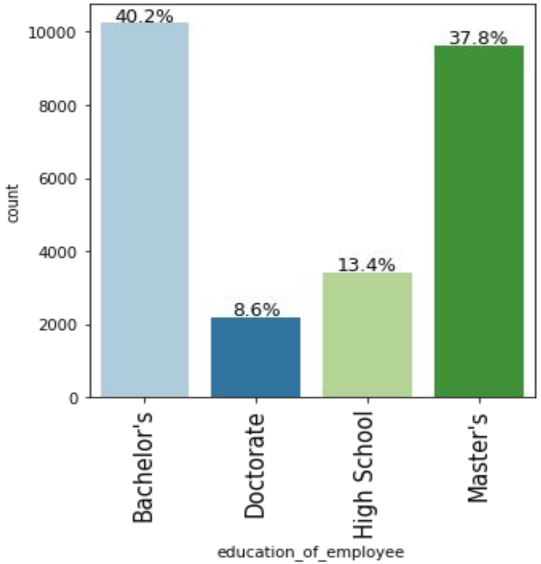
Business Problem Overview and Solution Approach

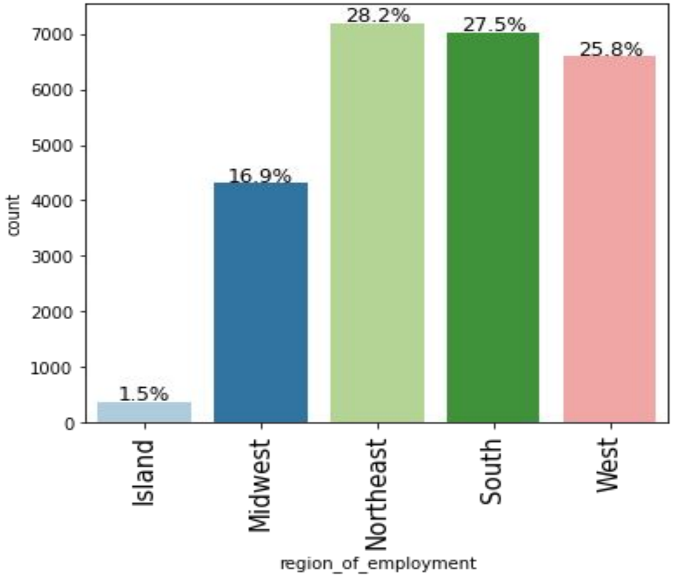
- OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

- The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year. In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications which was a nine percent increase in the overall number of processed applications from the previous year.

- The task at hand to analyze the data provided to facilitate the process of visa approvals by building a machine learning model and to recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Exploratory Data Analysis Results

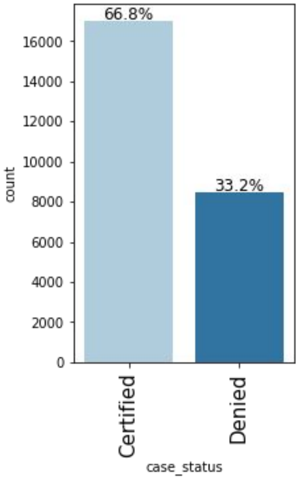


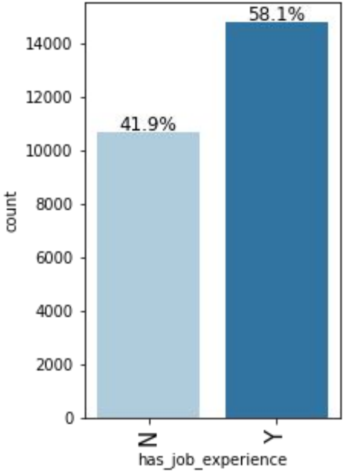


- 40.2% of the applicants have a bachelor's degree, followed by 37.8% having a master's degree.
- 8.6% of the applicants have a doctorate degree.

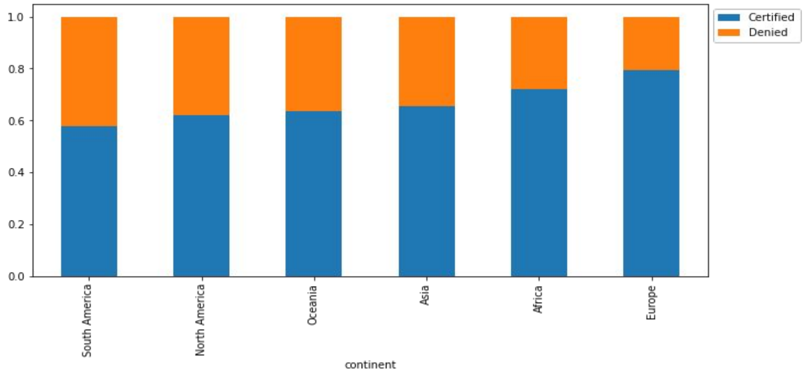
- Northeast, South, and West have almost equal percentages of applicants.
- The Island regions have only 1.5% of the applicants.

Exploratory Data Analysis



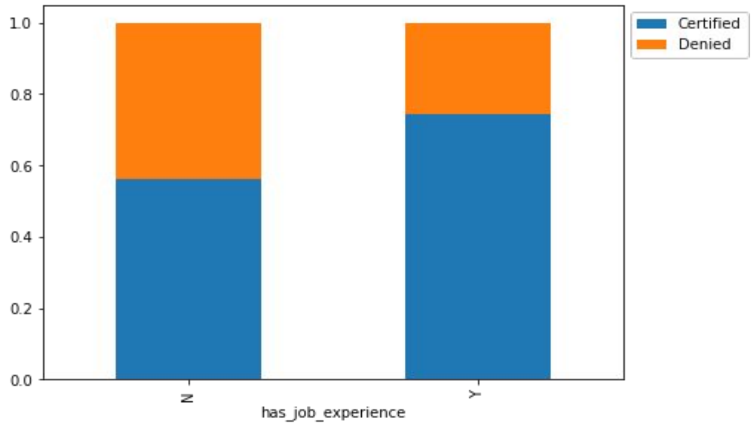


- 66.8% of the visas were certified
- 58.1% of the applicants have job experience.



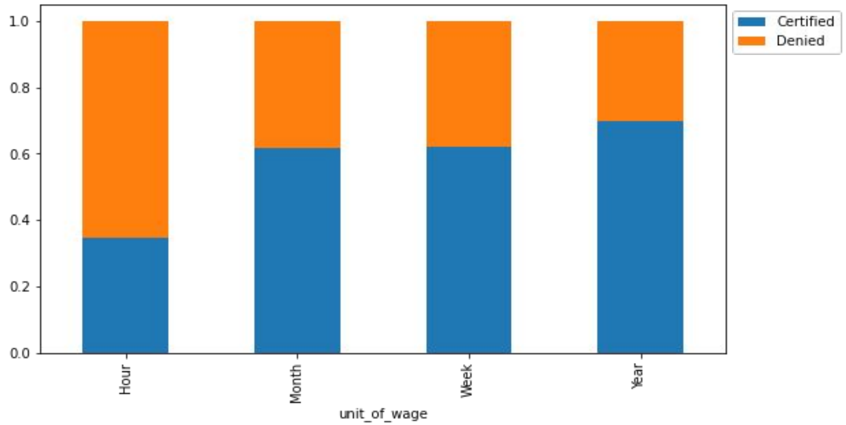
- Applications from **Europe** and **Africa** have a higher chance of getting certified.
- Around 80% of the applications from **Europe** are certified.
- **Asia** has the third-highest percentage (Around 60%) of visa certification and has the highest number of applications.

Exploratory Data Analysis

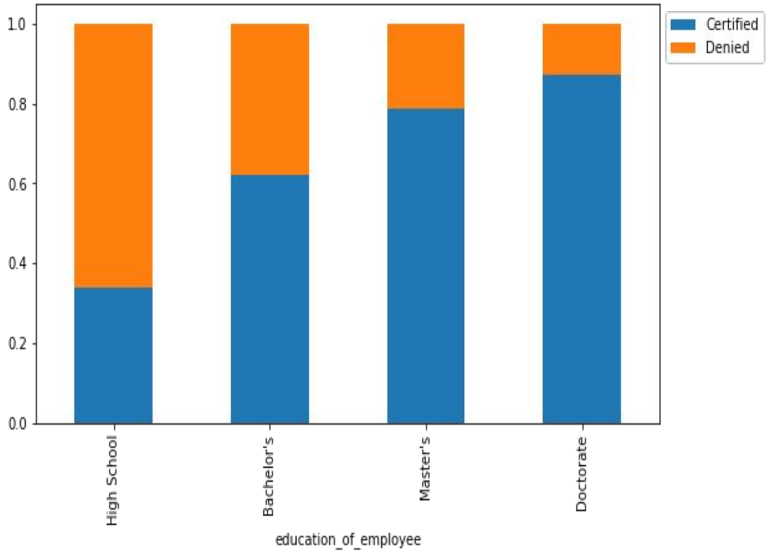


- **Having job experience** seems to be a key differentiator between visa applications getting certified or denied.
- Around 80% of the applications were certified for the applicants who have some job experience as compared to the applicants who do not have any job experience.
- Applicants without job experiences saw only 60% of the visa applications getting certified.

- **Unit of prevailing wage** is an important factor for differentiating between a certified and a denied visa application.
- If the unit of prevailing wage is Yearly, there's a high chance of the application getting certified.
- Around 75% of the applications were certified for the applicants who have a yearly unit of wage. While only 35% of the applications were certified for applicants who have an hourly unit of wage.

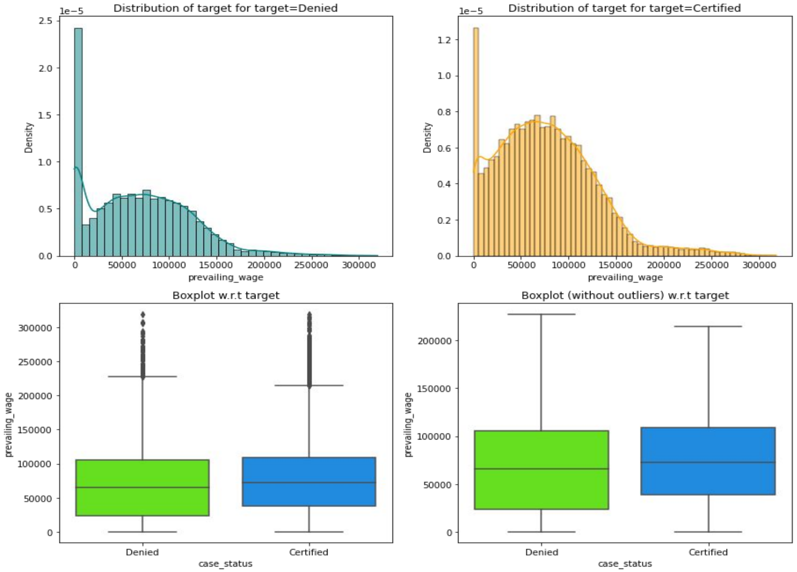


Exploratory Data Analysis

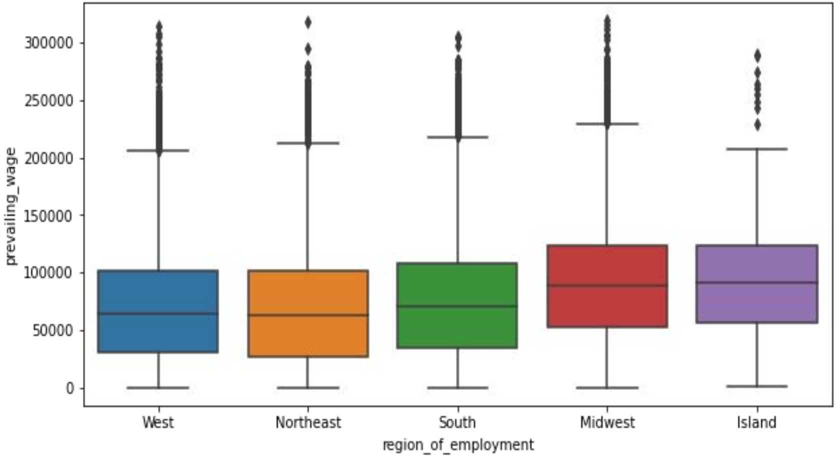


- **Education** seems to have a positive relationship with the certification of visa that is higher the education higher are the chances of visa getting certified.
- Around 85% of the visa applications got certified for the applicants with Doctorate degree. While 80% of the visa applications got certified for the applicants with Master's degree.
- Around 60% of the visa applications got certified for applicants with Bachelor's degrees.
- Applicants who do not have a degree and have graduated from high school are more likely to have their applications denied.

Exploratory Data Analysis



- The median prevailing wage for the **certified** applications is slightly higher as compared to **denied** applications.



- **Midwest** and **Island** regions have slightly higher prevailing wages as compared to other regions.
- The distribution of prevailing wage is similar across **West**, **Northeast**, and **South regions**.

Data Preprocessing

- There were no null and duplicate values in the dataset.
- The case_id column was dropped from the dataset since all the values in that column were unique.
- There were some negative values in the number of employees column which were replaced by their absolute values.
- There were quite a few outliers in the data, however, they were not treated since they were genuine values.
- The dataset is splitted into train,validation and test.
- Since the dataset is imbalanced, the data has been oversampled and undersampled.

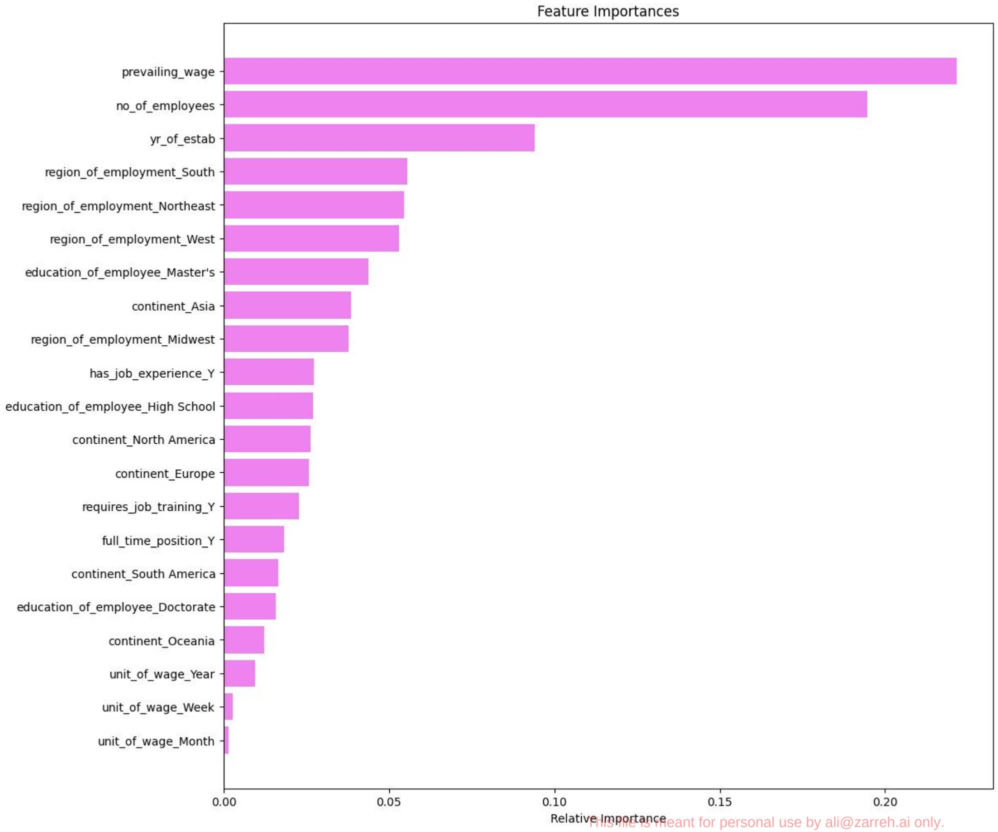
Model Performance Summary

- We want to predict whether the visa application will get certified or not using the information provided.
- We will use F1 Score as the metric for evaluation of the model because
    - If a visa is certified when it had to be denied a wrong employee will get the job position while US citizens will miss the opportunity to work on that position.
    - If a visa is denied when it had to be certified the U.S. will lose a suitable human resource that can contribute to the economy of the country.
    - F1 score will help us to minimize both false positives and false negatives.
- We will use **balanced class** weights so that model focuses equally on both classes.
- We will build different models - DecisionTreeClassifier, RandomForestClassifier, BaggingClassifier, AdaBoostClassifier, GradientBoostingClassifier, XGBClassifier, and StackingClassifier
- We will also perform **hyperparameter tuning** for these models and evaluate their performance using different metrics and confusion matrix.

Model Performance Summary

Model	Train Accuracy	Valid Accuracy	Train Recall	Valid Recall	Train Precision	Valid Precision	Train F1	Valid F1
Gradient Boost (Oversampled)	0.818140	0.707225	0.842021	0.799086	0.803637	0.770895	0.822382	0.784737
XGBoost (Oversampled)	0.815496	0.697194	0.999748	0.956465	0.730540	0.700016	0.844202	0.808389
AdaBoost (Oversampled)	0.813187	0.721180	0.831864	0.806704	0.801910	0.782517	0.816612	0.794427
Random Forest (Undersampled)	0.842141	0.699520	0.842985	0.709621	0.841564	0.816429	0.842274	0.759287

Feature Importance



The top **3 important features**

to look for while certifying a visa are -

- **Prevailing wage**
- **Number of Employees**
- **Year of establishment**

APPENDIX

Data Overview

- The data contains information of 25480 employees and their employers.
- The data includes information about employee's education, job experience, whether the employee requires training or not, continent, etc.
- The data also includes information about the number of employees in the company, year of establishment of the company, prevailing wage, etc.

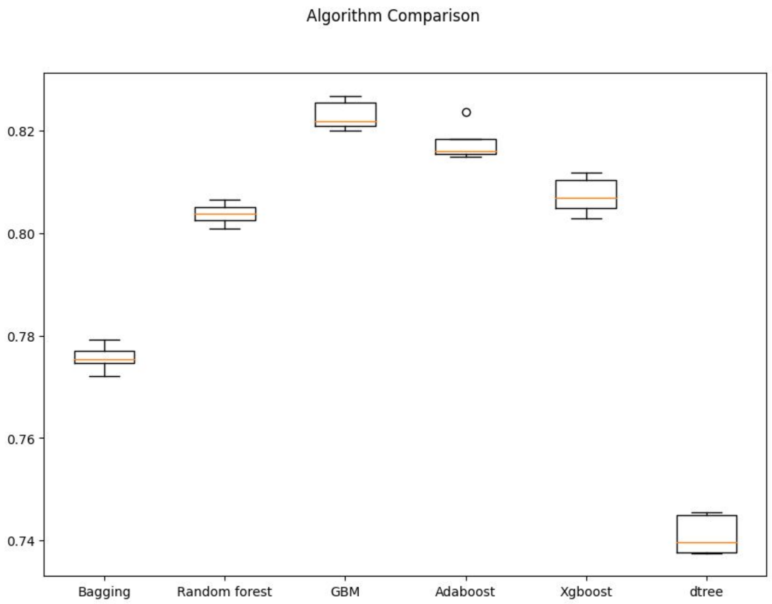
Models Performance on Original Data

```
Cross-Validation performance on training dataset:

Bagging: 0.7756586246579394
Random forest: 0.8037837241749051
GBM: 0.823039791269532
Adaboost: 0.8177228312991336
Xgboost: 0.8073583989766158
dtree: 0.7410652876513099

Validation Performance:

Bagging: 0.7675817565350541
Random forest: 0.7972364702187794
GBM: 0.8195818459969403
Adaboost: 0.8151868220168742
Xgboost: 0.8070320579110651
dtree: 0.7477497255762898
```



- F1 Score is highest for the GBM followed by AdaBoost, XGBoost and Random Forest.
- The variance of the scores is minimum in the case of Random Forest.

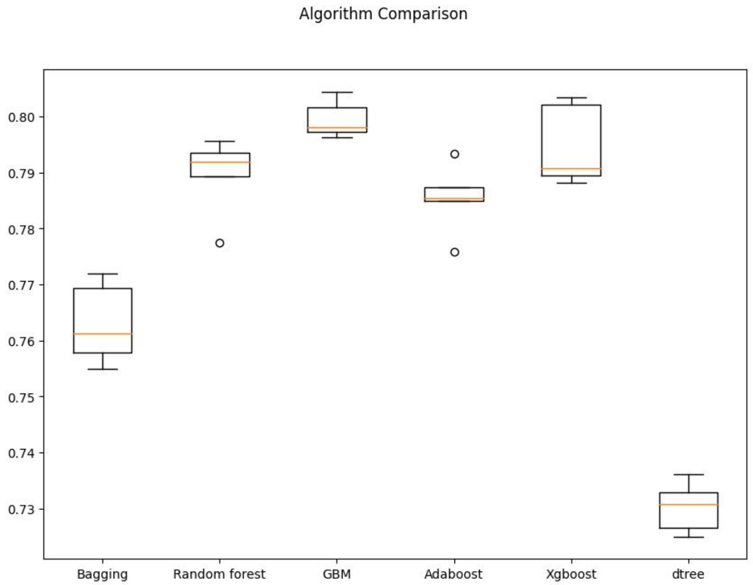
Models Performance on Over Sampled Data

```
Cross-Validation performance on training dataset:

Bagging: 0.7630248470329939
Random forest: 0.789527508885066
GBM: 0.7995200290888033
Adaboost: 0.7853502765579534
Xgboost: 0.7947610182951765
dtree: 0.7302748433737385

Validation Performance:

Bagging: 0.7615769017212659
Random forest: 0.7873070325900514
GBM: 0.8013683985460766
Adaboost: 0.7846414216204781
Xgboost: 0.7933446375582874
dtree: 0.7402482269503546
```



- The F1 score is highest for GBM, followed by AdaBoost, XGBoost, and Random Forest.
- The variance of the scores for Random Forest, GBM, and AdaBoost is similar.

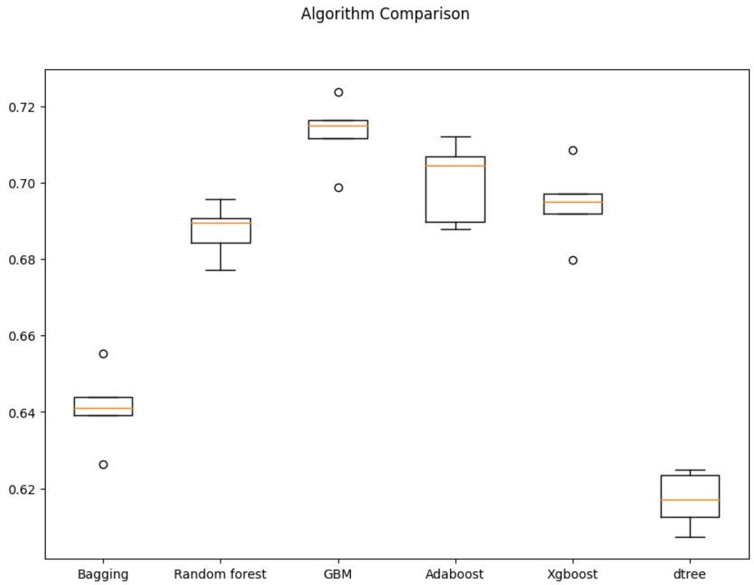
Models Performance on Under Sampled Data


```
Cross-Validation performance on training dataset:

Bagging: 0.6411413525524321
Random forest: 0.6875011408129813
GBM: 0.7131358906535971
Adaboost: 0.7002423692713601
Xgboost: 0.6944693136408734
dtree: 0.617022679979161

Validation Performance:

Bagging: 0.6916956737941323
Random forest: 0.734144015259895
GBM: 0.7608695652173914
Adaboost: 0.7559219693450998
Xgboost: 0.7423652871123688
dtree: 0.6839080459770115
```



- The F1 score is highest for GBM, followed by AdaBoost, XGBoost, and Random Forest.
- The variance of the scores for AdaBoost has increased.

Hyperparameter Tuning - AdaBoost (Oversampled)

	Accuracy	Recall	Precision	F1
0	0.813187	0.831864	0.80191	0.816612

Training Performance

	Accuracy	Recall	Precision	F1
0	0.72118	0.806704	0.782517	0.794427

Validation Performance

- AdaBoost is slightly overfitting the train data but overall the performance has improved.

Hyperparameter Tuning - Random Forest (Undersampled)

	Accuracy	Recall	Precision	F1
0	0.842141	0.842985	0.841564	0.842274

Training Performance

	Accuracy	Recall	Precision	F1
0	0.69952	0.709621	0.816429	0.759287

Validation Performance

- Random forest classifier is slightly overfitting the train data but overall the performance has improved.

Hyperparameter Tuning - Gradient Boost (Oversampled)

	Accuracy	Recall	Precision	F1
0	0.81814	0.842021	0.803637	0.822382

Training Performance

	Accuracy	Recall	Precision	F1
0	0.707225	0.799086	0.770895	0.784737

Validation Performance

- Gradient Boosting is slightly overfitting the train data but overall the performance has improved.

Hyperparameter Tuning - XGBoost (Oversampled)

	Accuracy	Recall	Precision	F1
0	0.815496	0.999748	0.73054	0.844202

Training Performance

	Accuracy	Recall	Precision	F1
0	0.697194	0.956465	0.700016	0.808389

Validation Performance

- Overfitting has further increased using oversampled data and the model is performing good on train data.

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Power Ahead

Happy Learning !